

ROS2 Network Issues & Solution

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Issue Description

When running ROS2 across multiple machines for the Assembly Line Project, we observed the following;

1. Inconsistent message delivery: Laptop_1 can publish messages, but Laptop_2, Laptop_3, Laptop_n sometimes miss messages sporadically.
2. Unpredicted latency spikes: Even though I set the QoS across all devices to be Reliable, there is still delays if the message eventually arrives.
3. Pattern unclear: Sometimes it works well, and fails other times.

Possible Causes

After running network diagnostics and analyzing packets of the various ROS2 devices. I suspect that the following might be potential issues:

1. QoS mismatch: I ran multiple tests on various ROS2 devices and I realized that having a QoS mismatch on the devices (both the publisher and subscriber) can cause the issue.
2. Network Issue: I scanned the network to check for packet loss, and I exposed the various ROS2 ports (range of 32650, *) to ensure the firewall policies is non-blocking.
3. DDS Middleware: Unlike ROS1 that solely relies on TCP, ROS2 relies on both TCP and UDP(primarily), this brings flexibility but it has its cons. For instance, making use of best effort (udp) might result to packet loss during communication.
4. Multicast Problems: I suspect there might be an issue in the Router settings, my guess is to enable QoS on the Router and set the priority of service, interface, and set netmask and mac address as a priority. So that when the network is stressed with many devices, it knows what to prioritize.

What I have tried

Below are some of the things I have tried;

Test	Command	Result
Reliable QoS	<code>ros2 topic pub -qos-reliability reliable</code>	Still occasional drops
Best Effort QoS	<code>ros2 topic pub -qos-reliability best_effort</code>	More drops, but lower latency
Transient Local Durability	<code>-qos-durability transient_local</code>	This helped for late-joining subscribers
Increased Depth	<code>-qos-depth 50</code>	Reduced but didn't eliminate drops

Network Diagnostics

Test	Command	Result
Packet Loss Simulation	<code>sudo tc qdisc add dev lo root netem loss 5%</code>	I confirmed that QoS reliable helps but isn't perfect, once I increased the packet loss, issue remained
Firewall CHeck	<code>ufw disable</code>	No significant difference
Router QoS	Enabled/Disabled	Have not ran extensive test on this
IGMP Version (v2)	Checked router settings	Multicast worked but may not be optimal, have not ran extensive test
ROS2 Packet Loss Measure	<code>ros2 topic hz</code>	Discovered loss in packet and delayed time of arrival of data
FLow Control	Enabled Tx/Rx	No clear impact at the time of writing, yet to run extensive test

DDS Middleware

Middleware	Command	Result
Fast DDS	<code>export RMW_IMPLEMENTATION=rmw_fastrtps_cpp</code>	Slightly better reliability
Cyclone DDS	<code>export RMW_IMPLEMENTATION=rmw_cyclonedds_cpp</code>	A bit more consistent than Fast DDS

Conclusion

1. ROS2 QoS alone is not enough if the network has issues
2. Best results I observed came from the following settings;
 - Wired connection
 - Matching QoS settings.
 - Proper DDS tuning (multicast for better result)

To-Do

- Stress Test with 5 Raspberry Pis; To validate the ROS2 Performance under high network load
- Testing for both ROS2 middleware (Fast DDS & Cyclone DDS)